Introduction to IoT

Question Bank

- Define the Internet of Things (IoT) and list any four of its key characteristics.
- What are the main components of an IoT system?
- What are the advantages of IoT?
- What is cloud computing in IoT?
- What is the role of a microcontroller in an IoT project?
- Differentiate between analog and digital sensors.
- What is edge computing in IoT?
- Differentiate between IoT and M2M communication.
- What are “Things” in the context of IoT?
- Briefly discuss the key features of IoT.
- Explain the meaning of “Anything Anywhere Anytime” in the context of IoT.
- Explain the evolution of M2M to IoT.
- Write a short note on interoperability and scalability in IoT design.
- Explain the evolution of IoT, highlighting key milestones and technological developments.
- Discuss the enabling technologies used in IoT and how they contribute to efficient IoT system design.
- Describe the characteristics, benefits, and challenges of IoT in detail.
- Discuss in detail some of the major application domains of IoT in today’s world. What are the common threats, and how can they be mitigated?
- Describe the role of an actuator in an IoT system. Also give two examples of actuators commonly used in IoT applications.
- What is a Piezoelectric sensor?
- Explain the function of the perception/sensing layer in IoT architecture.
- Is a touch screen a sensor or an actuator? Justify your answer.
- Describe different types of sensors commonly used to sense temperature and light intensity.
- Explain the key principles that should be followed while designing an Internet of Things (IoT) system.
- Briefly describe the five-layer architecture of IoT.
- What is the role of sensors and actuators in IoT?
- List and explain any three IoT enabling technologies.
- What is the purpose of middleware in an IoT ecosystem?
- Discuss the important design principles of IoT systems with suitable examples.
- Explain the major differences between active and passive sensors with suitable examples for each type.
- Explain the working principles of Infrared (IR), Accelerometer and Gyroscope sensors.
- Describe the architecture and functions of embedded systems in IoT.
- Describe the role and working of RFID technology in IoT.
- Name two long-range communication technologies used in IoT and mention one use case for each.
- Describe the working principle of Near Field Communication (NFC) technology. List out some real-life applications of NFC.

- Compare the communication models: Device-to-Device (D2D), Device-to-Cloud (D2C), and Device-to-Gateway (D2G) in terms of latency, internet dependency, and data processing location.
- Differentiate between Zigbee and Bluetooth technologies in the context of IoT.
- Write a short note on MQTT protocol and its relevance in IoT.
- Explain the Device-to-Gateway communication model with an example.
- Explain the process of collecting sensor data using a Zigbee network with suitable diagram.
- Explain Piconet and Scatternet architecture of Bluetooth technology.
- Write short notes on MQTT and HTTP.
- Compare and contrast CoAP and MQTT protocols used in IoT.
- Explain various IoT communication models with real-world examples.
- Discuss four commonly used data processing techniques in IoT with proper examples.
- Discuss two major privacy concerns in IoT deployments.
- Discuss the 5Vs of Big Data in IoT.
- Explain IoT data life cycle with proper diagram.
- Compare and contrast the three types of data storage approaches used in IoT: Cloud, Edge, and Fog Computing.
- Discuss in detail the security threats and mitigation strategies in IoT.
- Describe the role of cloud computing in IoT with advantages and limitations.
- Explain how AI and 5G are transforming future trends in IoT.
- Discuss the significance of energy management and sustainability in IoT deployments.
- Explain the role of middleware in IoT ecosystems.
- Describe different communication protocols used in IoT.
- Explain the significance of scalability in IoT systems.
- Explain how IoT is transforming the healthcare sector.
- Discuss the challenges faced in deploying IoT applications.
- Describe how IoT technology can be used to develop Smart City and Smart Farming applications.
- List three applications of IoT in agriculture.
- What are Digital Twins and how are they related to IoT?
- Explain the application of IoT in healthcare with suitable use cases.
- Discuss the Industrial IoT (IIoT) domain and its impact on manufacturing.
- Write a detailed note on IoT applications in smart cities.
- What are the challenges faced in deploying IoT systems? Suggest solutions.
- Explain future trends and emerging technologies in IoT.
- Discuss ethical and privacy issues related to IoT systems.
- Explain the impact of IoT on smart living and industrial automation.
- Describe the tasks of digitalWrite() and digitalRead().
- Explain the task of pinMode() in Arduino.
- Write a program for Arduino Uno to detect an obstacle using IR sensor.
- Write a program for Arduino Uno to measure temperature using DHT11 sensor